Hang Li

ACADEMIC CURRICULUM VITAE

Postdoctoral Research Fellow | Information Retrieval and Artificial Intelligence
The University of Queensland, Brisbane, Australia

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ACADEMIC PROFILE

Postdoctoral Research Fellow at the University of Queensland specialising in information retrieval, large language models, and retrieval-augmented generation, with publications in SIGIR, WSDM, ECIR, and ACM Transactions on Information Systems. Teaching experience includes guest lecturer, and head teaching assistant roles. Principal supervisor of **four master's thesis students** and associate supervisor of **one PhD student** (another one incoming). Combines research in neural retrieval, agentic search, and relevance evaluation with applied software engineering experience.

EDUCATION

PhD in Computing Science — April 2020 – February 2026
The University of Queensland — Brisbane, Australia

- Thesis: *Advanced Query Representation and Feedback Methods for Neural Information Retrieval*.
- Supervisors: Prof. Guido Zuccon (principal) and A/Prof. Bevan Koopman.
- Research supported by the AgAsk project (UOQ2003-009RTX).

BSc in Computer Science — September 2012 – July 2016
University of Minnesota, Twin Cities — Minneapolis, United States

- Global Excellence Scholarship, 2012–2016.

TEACHING EXPERIENCE

The University of Queensland: teaching roles in information retrieval, concurrency, and web information systems.

ROLE	COURSE	YEAR(S)
Guest Lecturer	INFS7410 Information Retrieval	2025–2026
Teaching Assistant	CSSE7610 Concurrency	2025
	INFS7410 Information Retrieval	2020–2024
Head Teaching Assistant	INFS7202 Web Information System	2020–2022

RESEARCH SUPERVISION

Master's Theses — Principal supervisor of **four master's thesis students**; all ongoing.

PhD Supervision — Associate supervisor of **one PhD student** (another one incoming), with a **40% supervision contribution per student**; ongoing. Provide day-to-day research supervision.

Guide students in research methodology, experimental design, and software development across information retrieval, machine learning, and large language models. Also mentor undergraduate and honours research students.

ACADEMIC SERVICE

Conference Organisation — Proceedings Chair, CIKM 2021

Program Committees — SIGIR, CIKM, ECIR, WSDM, and The Web Conference

Peer Review — ACM Transactions on Information Systems; ACL Rolling Review

ACADEMIC AND RESEARCH EXPERIENCE

Postdoctoral Research Fellow October 2025 – Present
Information Engineering Lab (IELab), The University of Queensland Brisbane, Australia

- Develop retrieval-augmented generation systems for clinical question answering, spanning data curation, model fine-tuning, retrieval pipelines, cloud deployment, and research prototypes.
- Investigate LLM-based relevance assessment, including overrating by LLM judges and the reliability and cost of automated evaluation.
- Research agentic search and LLM-based retrieval and reranking; supervise postgraduate research and contribute to collaborative publications.
- Contribute to research funding applications and project reports.

Research Assistant / PhD Researcher December 2020 – February 2026
IELab, The University of Queensland Brisbane, Australia

- Developed and evaluated pseudo-relevance feedback methods for neural retrieval, studying query representations, feedback quality, and the combination of sparse and dense retrievers.
- Contributed to AgAsk, a conversational search system for agricultural questions, and Agvaluate, a test collection for agricultural passage and document retrieval.
- Published research in SIGIR, WSDM, ECIR, and TOIS; released research code and datasets to support reproducibility.
- Mentored student research projects, presented at international conferences, and contributed to project reports and research funding proposals.

Summer Research Scholar November 2019 – March 2020
IELab, The University of Queensland Brisbane, Australia

- Developed search and recommendation prototypes for systematic review automation, contributing to research on end-to-end query formulation published at SIGIR 2020.

INDUSTRY EXPERIENCE

Software engineering experience in telehealth, e-commerce, financial technology, and enterprise IT asset management, covering backend services, RESTful APIs, databases, and cloud deployment.

Full Stack Developer July 2018 – July 2019
Henan She Chuan Technology Co., Ltd. Zhengzhou, China

Backend Developer June 2017 – March 2018
Shenzhen Dianmao Technology Group Co., Ltd. Shenzhen, China

Full Stack Developer August 2016 – June 2017
First Capital Fund Management Co., Ltd. Shenzhen, China

Software Developer February 2016 – July 2016
The Velocity Tech Solutions Inc. Minnesota, United States

RESEARCH GRANT APPLICATIONS

2026 MRFF Early to Mid-Career Researcher Grant
Co-design and preliminary testing of an AI-enabled tool for personalised dementia risk reduction in Parkinson's disease (PRISM-PD)
Role: Chief Investigator **Funding requested:** AUD 522,384.30 **Status:** Submitted; pending review.

AWARDS AND SCHOLARSHIPS

2022 **SIGIR Student Travel Grant**
2021 **Huawei DIGIX Global AI Challenge**; Team global champion; US$30,000 prize
2021 **ADCS Best Student Paper Award**
2020 **SIGIR Student Travel Grant**
2020 **UQ Research Training Program Stipend**; Four-year award
2019 **UQ Dean's Commendation for Academic Excellence**
2012 **Global Excellence Scholarship**; Four-year award

PUBLICATIONS

Journal Articles

J1 **AgAsk: An Agent to Help Answer Farmer's Questions from Scientific Documents**
Bevan Koopman, Ahmed Mourad, **Hang Li**, Anton van der Vegt, Shengyao Zhuang, Simon Gibson, Yash Dang, David Lawrence, Guido Zuccon
International Journal on Digital Libraries 25(4), pp. 569–584, 2024.

J2 **Pseudo Relevance Feedback with Deep Language Models and Dense Retrievers: Successes and Pitfalls**
Hang Li, Ahmed Mourad, Shengyao Zhuang, Bevan Koopman, Guido Zuccon
ACM Transactions on Information Systems 41(3), pp. 1–40, 2023.

J3 **Design and Research of Intelligent Question-Answering (Q&A) System Based on High School Course Knowledge Graph**
Zhijun Yang, Yang Wang, Jianhou Gan, **Hang Li**, Ning Lei
Mobile Networks and Applications, pp. 1–7, 2021.

Conference Papers

C1 **Evalugator—Rapid, Agile Development and Evaluation of Retrieval Augmented Generation Systems Without Labels**
Bevan Koopman, **Hang Li**, Shuai Wang, Guido Zuccon
ECIR 2026, pp. 63–67.

C2 **When LLM Judges Inflate Scores: Exploring Overrating in Relevance Assessment**
Chuting Yu, **Hang Li**, Guido Zuccon, Joel Mackenzie, Teerapong Leelanupab
ACM SIGIR 2026, pp. 4322–4327.

C3 **AgAsk: A Conversational Search Agent for Answering Agricultural Questions**
Hang Li, Bevan Koopman, Ahmed Mourad, Guido Zuccon
ACM WSDM 2023, pp. 1140–1143.

C4 **Mesh Suggester: A Library and System for MeSH Term Suggestion for Systematic Review Boolean Query Construction**
Shuai Wang, **Hang Li**, Guido Zuccon
ACM WSDM 2023, pp. 1176–1179.

C5 **How Does Feedback Signal Quality Impact Effectiveness of Pseudo Relevance Feedback for Passage Retrieval**
Hang Li, Ahmed Mourad, Bevan Koopman, Guido Zuccon
ACM SIGIR 2022, pp. 2154–2158.

C6 **To Interpolate or Not to Interpolate: PRF, Dense and Sparse Retrievers**
Hang Li, Shuai Wang, Shengyao Zhuang, Ahmed Mourad, Xueguang Ma, Jimmy Lin, Guido Zuccon
ACM SIGIR 2022, pp. 2495–2500.

C7 **Pseudo-Relevance Feedback with Dense Retrievers in Pyserini**
Hang Li, Shengyao Zhuang, Xueguang Ma, Jimmy Lin, Guido Zuccon
ADCS 2022, pp. 1–6.

C8 **Improving Query Representations for Dense Retrieval with Pseudo Relevance Feedback: A Reproducibility Study**
Hang Li, Shengyao Zhuang, Ahmed Mourad, Xueguang Ma, Jimmy Lin, Guido Zuccon
ECIR 2022, pp. 599–612.

C9 **Implicit Feedback for Dense Passage Retrieval: A Counterfactual Approach**
Shengyao Zhuang, **Hang Li**, Guido Zuccon
ACM SIGIR 2022, pp. 18–28.

C10 **MeSH Term Suggestion for Systematic Review Literature Search**
Shuai Wang, **Hang Li**, Harrisen Scells, Daniel Locke, Guido Zuccon
ADCS 2021, pp. 1–8.

C11 **Deep Query Likelihood Model for Information Retrieval**
Shengyao Zhuang, **Hang Li**, Guido Zuccon
ECIR 2021, pp. 463–470.

C12 **Systematic Review Automation Tools for End-to-End Query Formulation**
Hang Li, Harrisen Scells, Guido Zuccon
ACM SIGIR 2020, pp. 2141–2144.

TREC Track Papers

T1 **IELAB at TREC Deep Learning Track 2021**
Shengyao Zhuang, **Hang Li**, Shuai Wang, Guido Zuccon
Proceedings of the 2021 TREC Deep Learning Track, 2021.

T2 **IELAB for TREC Conversational Assistance Track (CAsT) 2020**
Sebastian Cross, **Hang Li**, Arvin Zhuang, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Proceedings of the 29th Text REtrieval Conference (TREC 2020), 2020.

Datasets

D1 **Agvaluate: A New Test Collection for Both Passage and Document Retrieval in the Agriculture Domain**
Hang Li, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Research dataset, The University of Queensland, 2022.

Preprints and Submitted Manuscripts

P1 **Effective Dense Retrieval using Only In-Context Examples**
Nour Jedidi, Abdul Basit Ali, **Hang Li**, Jimmy Lin
Submitted to the 2027 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), 2026.

P2 **AOBA: Adaptive Overlapping Block Aggregation for Efficient LLM Reranking**
Hang Li, Bevan Koopman, Guido Zuccon
Submitted to the 49th European Conference on Information Retrieval (ECIR), 2026.

P3 **Whole-Pool Setwise Reranking with Long-Context Language Models**
Hang Li, Chuting Yu, Teerapong Leelanupab, Bevan Koopman, Guido Zuccon
Submitted to the 49th European Conference on Information Retrieval (ECIR), 2026.

P4 **Bridging the Gap Between Small and Large Dense Retrievers via Offline Generative Feedback**
Hang Li, Chuting Yu, Xiao Wang, Bevan Koopman, Guido Zuccon
Submitted to ACM Transactions on Information Systems, 2026.

P5 **A Survey of Late-Interaction Neural Retrieval: Paradigms, Systems, and Research Frontiers**
Xiao Wang, Chuting Yu, Minghan Li, Binci Yang, **Hang Li**, Ben He
Submitted to Information Sciences, 2026.

P6 **Pseudo-Relevance Feedback Can Improve Zero-Shot LLM-Based Dense Retrieval**
Hang Li, Xiao Wang, Bevan Koopman, Guido Zuccon
Preprint, 2025. arXiv: `2503.14887v1 [cs.IR]`.

P7 **LLM-VPRF: Large Language Model Based Vector Pseudo Relevance Feedback**
Hang Li, Shengyao Zhuang, Bevan Koopman, Guido Zuccon
Preprint, 2025. arXiv: `2504.01448 [cs.IR]`.

P8 **TPRF: A Transformer-based Pseudo-Relevance Feedback Model for Efficient and Effective Retrieval**
Hang Li, Chuting Yu, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Preprint, 2024. arXiv: `2401.13509 [cs.IR]`.

TECHNICAL AND RESEARCH SKILLS

Research Methods	Retrieval evaluation; reproducible experimentation; query representation and feedback; LLM-based relevance assessment; retrieval-augmented generation
Programming and ML	Python, Java, C++, PyTorch, Hugging Face Transformers, PyTerrier, Pyserini, Pandas, NumPy
Systems and Tools	Docker, OpenShift, AWS, Linux, Git, MySQL, MongoDB, Redis, LaTeX